Enterprise Agentic AI Banking Platform Architecture

Version 1.0 (Baseline)

Document Status: Fixed Baseline (Version 1.0)

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1. Objective

Build an enterprise-grade Agentic AI Banking Platform capable of orchestrating banking capabilities using AI Agents, MCP (Model Context Protocol), and durable workflow orchestration while preserving domain ownership and keeping Core Banking as the system of record.

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2. Architecture Vision

The platform is built around the following principles:

- AI Agents understand customer intent and make business decisions.
- MCP standardizes capability discovery and invocation.
- Workflow Runtime provides durable execution for long-running business processes.
- Business domains own their capabilities and workflows.
- Business Services implement business logic.
- Core Banking remains the financial system of record.

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3. High-Level Platform Architecture

```
                    Customer
                       │
                       ▼
         Channels (Web / Mobile / API)
                       │
                       ▼
          Conversation / API Layer
                       │
                       ▼
            AI Agent Orchestration
      (Planner • Coordinator • Reasoning)
                       │
        ┌──────────────┴──────────────┐
        │                             │
```

```
              ▼                              ▼
       LLM Connectivity              Enterprise Memory
    (OpenAI / Azure OpenAI)      (Conversation Context)
              │                              │
              └──────────────┬───────────────┘
                             │
                             ▼
                        MCP Client
                             │
        ┌────────────────────┴────────────────────────┐
        │                                             │
        ▼                                             ▼
  Enterprise MCP Servers                      Domain MCP Servers
 (Cross-Domain Capabilities)           (Single Domain Capabilities)
        │                                             │
        └────────────────────┬────────────────────────┘
                             ▼
                Workflow Runtime (Temporal)
                             │
                    Temporal Workers
                             │
              Domain Workflows & Activities
                             │
                     Business Services
                             │
                             ▼
             Core Banking / Enterprise Systems
```

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4. Agentic AI Banking Architecture Patterns

Pattern A – Direct Capability Invocation

Best suited when

- Single business capability
- Immediate synchronous response
- No workflow coordination required

Not suited when

- Long-running processing
- Cross-domain orchestration
- Compensation or retries are required

Architecture

```
Customer
    │
    ▼
AI Agent
    │
    ▼
LLM
    │
    ▼
MCP Client
    │
    ▼
Domain MCP Server
    │
    ▼
Business Service
    │
    ▼
Core Banking
```

Banking Examples

- Balance Inquiry
- Customer Profile
- Exchange Rate
- Transaction History

Pattern B – Capability-Owned Workflow

Best suited when

- A single business capability requires reliable workflow execution.
- The complete process belongs to one business capability.

Not suited when

- Multiple business domains participate.
- Enterprise-wide orchestration is required.

Architecture

```
Customer
    │
    ▼
AI Agent
```

```
     │
     ▼
LLM
     │
     ▼
MCP Client
     │
     ▼
Domain MCP Server
     │
     ▼
Workflow Runtime
   (Temporal)
     │
     ▼
Worker
     │
     ▼
Domain Workflow
     │
     ▼
Business Service
     │
     ▼
Core Banking
```

Banking Examples

- Fund Transfer
- Bill Payment
- Card Replacement
- Fixed Deposit Creation

Pattern C – Domain Workflow Orchestration

Best suited when

- Multiple activities exist within one business domain.
- A domain owns its complete orchestration.

Not suited when

- Multiple business domains must participate.

Architecture

```
Customer
     │
     ▼
AI Agent
     │
     ▼
LLM
     │
     ▼
MCP Client
     │
     ▼
Payment MCP Server
     │
     ▼
Workflow Runtime
   (Temporal)
     │
     ▼
Payment Workflow
     │
  ┌──┼──────────────┐
  ▼  ▼              ▼

Debit
Fraud Check
Ledger
     │
     ▼
Payment Service
     │
     ▼
Core Banking
```

Banking Examples

- Payment Processing
- Treasury Processing
- FX Processing
- Tax Processing

Pattern D – Enterprise Process Orchestration

Best suited when

- Multiple business domains participate.
- Enterprise business processes require centralized orchestration.

Not suited when

- A single domain can complete the request independently.

Architecture

```
Customer
     │
     ▼
AI Agent
     │
     ▼
LLM
     │
     ▼
MCP Client
     │
     ▼
Enterprise MCP Server
     │
     ▼
Workflow Runtime
   (Temporal)
     │
  ┌──┼───────────────┐
  ▼  ▼               ▼

Payment Workflow
Treasury Workflow
Tax Workflow
     │
     ▼
Business Services
     │
     ▼
Core Banking
```

Banking Examples

- Corporate Payments
- Customer Onboarding

- Loan Origination
- Trade Finance

Pattern E – Hybrid Enterprise Agentic Architecture (Recommended)

Best suited when

- The platform supports both simple capabilities and enterprise business processes.
- AI Agents dynamically choose the appropriate execution model.

Not suited when

- Every request follows exactly the same execution model.

Architecture

```
                         Customer
                            │
                            ▼
                         AI Agent
                            │
                            ▼
                           LLM
                            │
                            ▼
                        MCP Client
                            │
             ┌──────────────┴──────────────┐
             ▼                             ▼

      Simple Capability              Business Process

             │                             │

     Domain MCP Server           Enterprise MCP Server

             │                             │

     Business Service            Workflow Runtime
                                    (Temporal)
                                        │
                     ┌──────────────────┼──────────────────┐
                     ▼                  ▼                  ▼

             Payment Workflow   Treasury Workflow   Tax Workflow
                     │                  │                  │
```

```
                         ▼                    ▼                    ▼
                 Business Services    Business Services    Business
Services
                         |                    |                    |
                         └────────────────────┼────────────────────┘
                                              ▼
                              Core Banking / Enterprise Systems
```

Banking Examples

Direct Capability

- Balance Inquiry
- Customer Lookup
- Exchange Rate

Domain Workflow

- Fund Transfer
- Treasury Settlement
- Card Replacement

Enterprise Workflow

- Corporate Payments
- Loan Origination
- Customer Onboarding
- Cross-Border Payments

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5. Pattern Selection Matrix

Business Requirement	Recommended Pattern
Balance Inquiry	Pattern A
Single Capability requiring Workflow	Pattern B
Domain Process	Pattern C
Cross-Domain Business Process	Pattern D
Enterprise Banking Platform	Pattern E

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6. Technology Mapping (Version 1.0)

Architecture Layer	Responsibility	Version 1.0 Technology
Channels	Customer interaction	Web, Mobile, REST APIs

Architecture Layer	Responsibility	Version 1.0 Technology
Conversation Layer	Session and request handling	Spring Boot
AI Agent	Planning, reasoning and orchestration	Agent SDK
LLM Connectivity	Natural language reasoning	OpenAI / Azure OpenAI
Enterprise Memory	Conversation context	Agent Memory (Optional)
MCP Client	Tool discovery and invocation	MCP Client SDK
Enterprise MCP Server	Cross-domain capabilities	Spring Boot + MCP
Domain MCP Server	Domain capabilities	Spring Boot + MCP
Workflow Runtime	Durable workflow execution	Temporal Server
Workflow Processor	Workflow & Activity execution	Temporal Worker
Business Services	Banking capabilities	Spring Boot Microservices
Core Banking	Financial system of record	Core Banking Platform
External Systems	Third-party integration	REST, gRPC, MQ, ISO 20022, SWIFT

7. Architecture Principles (Version 1.0)

Principle	Description
AI-first Decision Making	AI Agents decide what business capability should be executed.
Capability-driven Architecture	Banking capabilities are exposed through MCP Servers.
MCP Standardization	AI Agents communicate only through MCP.
Workflow by Exception	Use workflow only when reliability, durability, retries, compensation, timers or long-running execution are required.
Enterprise before Domain	Cross-domain orchestration belongs to Enterprise MCP Servers.
Domain Ownership	Each domain owns its MCP Server, workflows, activities and business services.
Technology Independence	Business architecture remains independent of the workflow engine.
Separation of Responsibilities	AI Agents decide, MCP exposes capabilities, Workflow Runtime orchestrates, Business Services execute business logic.
Loose Coupling	Components evolve independently through well-defined contracts.

Principle	Description
Core Banking as System of Record	Financial truth always resides in Core Banking.
Independent Scalability	AI, MCP, Workflow Runtime and Business Services scale independently.
End-to-End Observability	Every request should be traceable across all platform layers.
Security by Design	Authentication, authorization and auditing apply across every layer.
Replaceable Technology	LLMs, Workflow Runtime and AI frameworks can be replaced with minimal business impact.

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8. Conclusion

The recommended target architecture for Version 1.0 is **Pattern E – Hybrid Enterprise Agentic Architecture**.

This architecture provides:

- AI-first business orchestration
- Standardized capability exposure through MCP
- Durable workflow execution using Temporal
- Clear enterprise and domain ownership
- Independent scalability
- Technology independence
- Enterprise-ready banking architecture

Version Status

Document: Enterprise Agentic AI Banking Platform Architecture

Version: 1.0

Status: Baseline Approved for Further Evolution

Future enhancements (v1.1, v2.0 and beyond) should extend this baseline while preserving its core architectural principles.